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MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT CKU-5/A/A ASSEMBLY

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for the procurement of the Rocket Catapult CKU-5/A/A loaded assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-S-9479	Seat System, Upward Ejection, Aircraft, General Specification for
MIL-P-82663	Propellant, Solid, Composite

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. N4546 FSC 1377

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MIL-STD-810 Environmental Test Methods and Engineering Guidelines

MIL-STD-1168 Ammunition Lot Numbering

MIL-STD-2101 Classification of Characteristics

MS 17983 Compass, Magnetic, Pilots Standby

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

DL5184322 List of Drawings, Catapult, Aircraft Ejections Seat, CKU-5/A/A Assembly

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E-165-80 Liquid Penetration Inspection Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specifiction, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified in the contract or order (see 6.2.1), first article samples of the catapult assemblies are required and shall be manufactured using the methods and procedures proposed for production. The samples shall be tested as specified in section 4 for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing catapult assemblies that comply with the technical requirements of the contract (see 4.4).

3.1.1 Pressure transducers. All first article samples catapult assemblies shall be adapted for pressure transducers as specified in the contract or order (see 6.2.1 and 6.5).

3.1.2 Propellant aging. Propellant samples shall be prepared and tested in accordance with the requirements of 4.4.3. Tests will be performed in accordance with MIL-P-82663. Failure of the propellant to meet the requirements of MIL-P-82663 will be cause for rejection of the batch.

3.2 Materials. All materials, parts, and assemblies shall be in accordance with the DL 5184322, this specification, and the documents referenced herein.

3.2.1 Primary Components. Only primary components from a single primary component lot shall be used in a production lot of loaded rocket catapults.

3.2.2 Propellant. The propellant shall be in accordance with MIL-P-82663.

3.3 Age.

3.3.1 Explosive components. All primers shall have been manufactured and certified within 24 months or recertified within 12 months of the assembly date of the catapult assembly lot. All other explosive components including but not limited to igniters, shall have been manufactured or recertified within 12 months of the assembly date for the rocket catapult assembly lot.

3.3.2 Propellant. The propellant shall be manufactured within the 150 days prior to the assembly date for the rocket catapult assembly lot.

3.4 Performance and product characteristics. The rocket catapults shall meet the following performance and product characteristics when test fired at −65, +70, and +165 °F using an ejected weight of 375 pounds.

3.4.1 Ballistic. The rocket catapults shall meet the ballistic performance parameters specified in table I.

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TABLE I. Ballistic performance requirements (375 pound ejected weight).

Parameter	−65 °F		70 °F		165 °F		AQL
	MIN	MAX	MIN	MAX	MIN	MAX	
Catapult							
Initiation pressure (psi)	400	600	400	600	400	600	—
Dynamic Response Index (DRI)[1]	—	18.0	—	18.0	—	22.0	0.65
Separation velocity (ft/s)[2]	35	—	35	—	35	—	1.0
Ignition delay (ms)	—	30	—	30	—	30	0.65
Axial thrust (lbf)	—	7000	—	7000	—	7000	1.0
Pressure (psi)	1200	—	1200	—	1200	—	1.0
Rocket motor							
Resultant impulse (lbf-s)	950	—	950	—	950	—	1.0
Ignition delay (ms)	—	30	—	30	—	30	1.0
Action time (ms)[3]	—	500	—	500	—	500	1.0
Resultant thrust (lbf)	—	5000	—	5000	—	5000	0.65
Thrust angle (reference)[4]	54 degrees		54 degrees		54 degrees		

[1]DRI is defined in MIL-S-9479 (see 6.3.7).
[2]Nominal separation velocities at 70 and 165 °F are 42.0 ft/s and 47.0 ft/s respectively.
[3]Nominal rocket motor action time at 165 °F is 350 ms and at −65 °F is 450 ms.
[4]Record for information only.

3.4.1.1 Other requirements. In addition to meeting the ballistic requirements, none of the following shall occur during test firing of rocket catapults:

a. No unit component, including but not limited to cartridge components, O-rings, springs, or pieces of propellant shall be ejected from the unit.

b. No component subject to loading shall fail.

c. No excessive nozzle erosion, hot spots, or burnthrough shall occur.

3.4.2 Initiation pressure. Maximum allowed initiation pressure input is 10,000 psi.

3.5 Hydrostatic pressure. The components listed in table II shall withstand the applicable internal hydrostatic pressure specified in table II for 15 seconds (s) minimum when tested in accordance with 4.5.6.2.

TABLE II. Hydrostatic pressure requirements.

Component	Drawing	Hydrostatic pressure (psi)
Motor tube	11726650	5,000
Launcher tube	11726655	12,000
Booster tube	11726656	12,000

3.6 Radiographic requirements. Each loaded rocket catapult shall be examined radiographically to ascertain the presence, condition, and correct placement of internal components.

3.7 Torque. The booster tube and breech assembly, when seated with a torque of 400 +50, −0 inch-pounds, shall not be loosened or deformed when subjected to a breakaway torque of 200 inch-pounds.

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3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The rocket catapult assembly, when passed in front of a MS17983 compass along the entire length of the catapult (see figure 1), shall not cause the compass indicator to deflect more than 5 degrees in either direction.

3.10 Environmental requirements. The rocket catapult assemblies shall withstand the environmental conditioning of 3.10.1 through 3.10.4. Rocket catapult assemblies subjected to this conditioning shall show no evidence of deterioration of propellant or disarrangement of components when examined in accordance with 4.5.6.1 and shall meet the ballistic performance requirements of 3.4 when test fired in accordance with 4.4.2.

3.10.1 High temperature. The rocket catapult shall withstand the high temperature test of 4.4.1.1.

3.10.2 Vibration. The rocket catapult shall withstand the vibration test of 4.4.1.2.

3.10.3 Temperature-shock. The rocket catapult shall withstand the temperature-shock test of 4.4.1.3.

3.10.4 Thirty-day storage test. The rocket catapult shall withstand the 30-day storage test of 4.4.1.4.

3.11 Identification markings. Each rocket catapult shall be marked in accordance with Drawing 5184322 with an identification label, a warning label, and an installation label.

3.11.1 Production lot designation. The production lot number shall follow MIL-STD-1168. The construction of the lot number shall be as follows:

XXX75D002-007
a b c d e

where

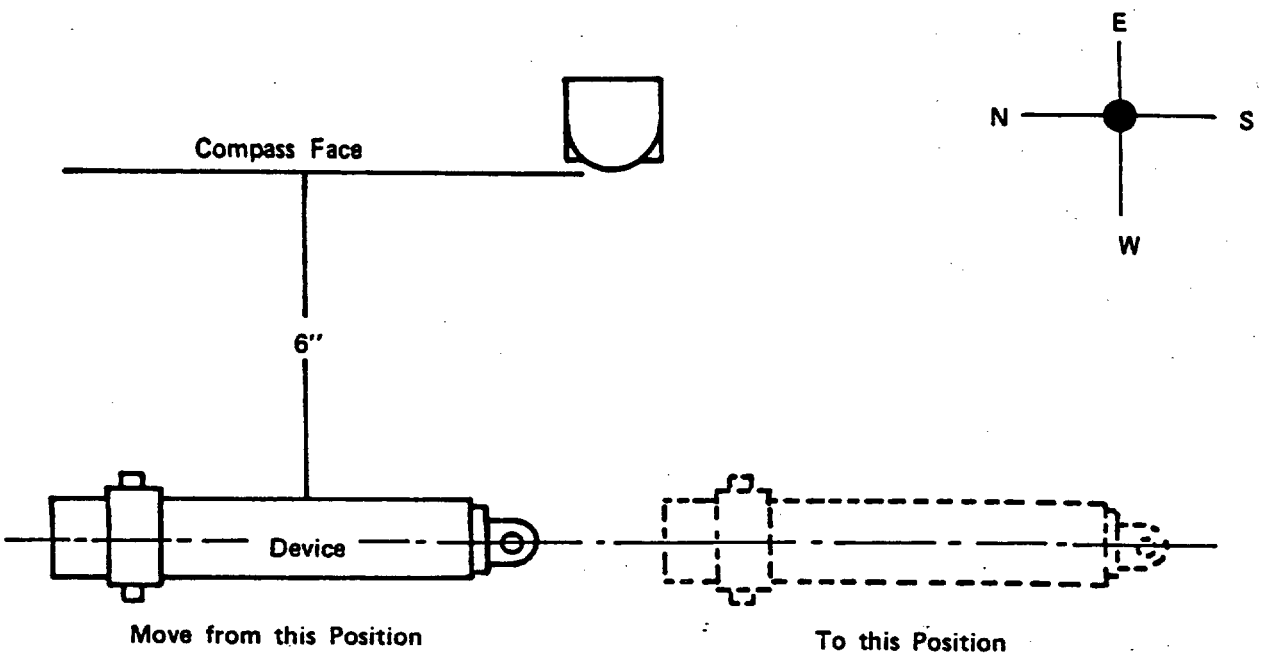
a = Manufacturer's identification code
b = The year of manufacture of the oldest batch of propellant (rocket motor or cartridge) in the lot
c = The month of manufacture of the oldest batch of propellant (rocket motor or cartridge) in the lot expressed as an alpha code in accordance with 4.1.3 of MIL-STD-1168
d = Lot interfix number (controlled by Naval Ordnance Station, Code 5120, Indian Head, MD 20640-5000 or contracting activity)
e = Lot sequence number - (manufacturer is responsible for the assignment of lot sequence numbers).

3.12 Workmanship. The workmanship shall be that required by the applicable drawings, referenced specifications, and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the unit or components, or be a safety hazard in handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.2.1).

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Note: Device and Compass Shall Lie in a Common Horizontal Plane

FIGURE 1. Layout for determination of residual magnetism in device.

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4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1) all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65 to 95 °F
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 First article inspection. Unless otherwise specified in the contract or order (see 6.2.1), a first article sample of 18 loaded rocket catapult assemblies, manufactured in accordance with 3.1, shall be submitted to the Naval Ordnance Station, Indian Head, Maryland 20640-5000 for first article inspection. The first article samples shall be subjected to the quality conformance inspection of 4.5, with the exception of the test firing of 4.5.5, followed by environmental testing, radiographic examinations, and static firing in the sequence specified in table III. Acceptance of the firt article sample shall be based on no defects in the sample. Any production of the rocket catapult assemblies by the contractor, prior to approval of the first article samples, shall be at the contractor's risk. First article samples shall not be included as a part of the quantity specified for delivery in the contract (see 6.2.1).

TABLE III. First article evaluation test program.[1]

Test Para.	Test	Unit number																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
4.5.6.1	Radiographic inspection	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a
4.4.1.1	High temperature	b	b	b	b	b	b												
4.4.1.2	Vibration −65 °F							b	b	b									
4.4.1.2	Vibration 165 °F										b	b	b						
4.4.1.4	30 day storage −65 °F													b	b	b			
4.4.1.4	30 day storage 165 °F																b	b	b
4.5.6.1	Radiographic inspection	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c
4.4.1.3	Temperature shock	d	d	d	d	d	d	d	d	d	d	d	d						
4.5.6.1	Radiographic inspection	e	e	e	e	e	e	e	e	e	e	e	e						
4.4.2	Ballistic test −65 °F	f	f					f			f			d			d		
4.4.2	Ballistic test 70 °F			f	f				f			f			d			d	
4.4.2	Ballistic test 165 °F					f	f			f			f			d			d

[1]Letters denote sequence of testing.

4.4.1 Environmental tests.

4.4.1.1 High temperature. The rocket catapult assembly will be subjected to the high temperature storage test in accordance with Method 501 of MIL-STD-810. The test items will be maintained at 165 °F for 50 hours minimum. Operation of the rocket catapult test sample is required only as specified in table III.

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4.4.1.2 Vibration. The rocket catapult assembly will be subjected to random vibration for jet aircraft in accordance with Method 514 of MIL-STD-810 for 6 hours with 2 hours on each axis. Before vibration, units will be temperature conditioned in accordance with table III for a minimum of 8 hours. The input vibratory acceleration must have a Wo factor of 0.1 g^2/Hz, and the reduction factor must be 0 g^2/Hz-dB. If the rocket catapult must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing will be stopped, and the unit will be reconditioned. A reconditioning time of 2 minutes for every minute the unit is outside the specified temperature envelope is required. A continuous temperature history of the unit will be kept for all conditioning and testing time. Unless otherwise noted, the rocket catapult must be mounted on a test fixture in a manner that will simulate actual seat/aircraft installation and excited through the attachment fittings. In addition the compliance with the random vibration test of Method 514, the rocket catapult shall comply with the gun fire vibration test of Method 519 for those applications whereby guns are installed on the aircraft (see 6.2.1). However, if the maximum test spectrum level of the gun fire configuration is determined to be equal to or less than the vibration test level of Method 514, the gun fire vibration test is not required.

4.4.1.3 Temperature shock. The rocket catapult assembly will be subjected to the temperature shock test in accordance with Method 503 of MIL-STD-810. The test temperature extremes will be −65 and +165 °F, and the duration of exposure at each temperature will be 8 hours. Operation of the rocket catapult test sample is required only as specified in table III.

4.4.1.4 Thirty-day storage. Three rocket catapults will be placed in storage at −65 °F and three at +165 °F for 30 days each. Insuring that the proper temperature is maintained is the only attention required outside normal working hours.

4.4.2 Static firing test. Within 30 days after completion of environmental exposure, the rocket catapults must be static fired at the Naval Ordnance Station, Indian Head, MD 20640-5000. The sample will be temperature conditioned per table III for a minimum of 8 hours and test fired within 10 minutes after removal from the conditioning chamber to verify conformance to the requirements of 3.4. Data specified in table I will be analyzed in accordance with MIL-STD-414, Section B, Part I, using the procedures of B-5 through B-7. Nonconformance with table I, or any requirement of 3.4.1, will be cause for rejection of the first article sample.

4.4.3 Propellant Aging. Three blocks of cured propellant from each batch of propellant used in the first article sample shall be submitted to the Naval Ordnance Station, Indian Head, Maryland 20640-5000, Attn: Code 5120, for aging testing. High temperature aging will be conducted for 84 days with an atmosphere of dry air and at 165 + 5 degrees Fahrenheit (°F). At zero time and at the end of 28-, 56-, and 84-day aging, physical properties shall conform to MIL-P-82663. The rate for stress/strain determination will be 2.0 inches/minute. The samples shall be cast into rectangular cardboard containers approximately 1.5 inches high by 8 inches wide by 11 inches long. All samples submitted shall be bare propellant, individually wrapped in moisture-proof packaging, and properly identified as to propellant lot and batch number.

4.4.4 Propellant morphology. Two motor tube and grain assemblies from each batch of propellant used in the first article sample are to be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000, Attn: Code 5120, for morphological examination of the propellant. This examination shall include microscopy and chromatography. All samples shall be individually wrapped in moisture-proof packaging and properly identified as to propellant lot, batch number, and grain serial number. These tests will be used as baseline information.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests of 4.5.4 through 4.5.6 conducted on samples selected as specified in 4.5.2. The contractor is responsible for assuring that all rocket catapults offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

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4.5.1 Test reports. When specified in the contract or order (see 6.2.1), the contractor shall furnish test reports (see 6.2.2) to the contracting activity. Test reports shall include the following information as described in 4.5.1.1 through 4.5.1.6:

a. Propellant description (see 3.2.2).
b. Static firing performance test results (see 4.5.5.1).
c. Summary sheet.
d. Ammunition data cards.
e. Propellant/assembly production records.
f. Certification of age of explosive components and propellant (see 3.3).

4.5.1.1 Propellant description. The following propellant information shall be submitted with each production lot (see 3.2.2). The materials and components of each propellant batch should be listed with the manufacturer, lot number, and date of manufacture. The burning rates at 1000, 2500, 3000, 4000, and 5000 psi performed at −65, 70, and 165 °F for each pressure (three samples at each temperature and pressure per mix) should also be included. In addition, include the results of tensile strength and elongation tests at −65, 70, and 165 °F with a pull rate of 2 inches/minute (three samples at each temperature per mix), heat of explosion (three samples per mix), and verification that ingredients were properly weighed and added to each mix (mix sheet (s)).

4.5.1.2 Static firing performance tests. All computations and data required for ballistics evaluation, generated during quality conformance inspection static firing per 4.5.5 (Government) and 4.5.5.1 (contractor) shall be reported including firing traces. The following information shall also be listed:

a. Grain designation, manufacture, lot number, and date of propellant manufacture
b. Type of test equipment
c. Deviations from standard testing procedures
d. Additional comments (malfunctions, unusual occurrences)
e. Disassembly findings report (see 4.5.5.2).

4.5.1.3 Summary sheet. A summary of the results of the tests for the parameters listed in table I shall be submitted to the designated Government activity for each inspection lot. The summary shall include the sample size, number and nature of defects found, and the disposition (accepted or rejected).

4.5.1.4 Data cards. Data cards (DD Form 1650) shall be completed for each loaded rocket catapult lot. Copies shall be provided in accordance with contract requirements, and one copy shall be forwarded to the Commanding Officer, Naval Ordnance Station, Indian Head, MD 20640-5000 (Code 5120).

4.5.1.5 Propellant/assembly production records. The contractor shall prepare and retain for the service life of the unit propellant production records including batch number, mix designation, rocket motor lot number, rocket motor serial number, cartridge lot number, and cartridge serial number. These shall be cross-referenced to the assembled rocket catapult lot number and to specific rocket catapult serial numbers.

4.5.1.6 Age certification. A certification of the age of explosive components and propellant shall be furnished by the contractor as required by 3.3.

4.5.2 Sampling for quality conformance inspection.

4.5.2.1 Lot. For the purposes of identification, inspection, and shipment, a lot shall consist of 420 loaded catapult assemblies, maximum, inclusive of test samples. The homogeneity of inspection lots shall be maintained as follows:

a. Components were produced by homogeneous process.
b. Units are loaded with same lot of rocket motors (see 6.3.10).

Only primary components from a single primary component lot shall be used in a production lot of loaded catapults.

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4.5.2.2 Sampling for examination. Designation of defects (critical, major, and minor) shall be in accordance with DOD-STD-2101.

a. Critical and major defects: Each rocket catapult asembly shall be examined for all critical and major defects.

b. Minor defects: Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.2.3 Sampling for firing test. A sample of loaded rocket catapult assemblies shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000, for static firing in accordance with 4.5.5. The sample size shall be in accordance with table IV. Samples must be chosen by Naval Ordnance Station, Indian Head, Maryland personnel or by a designated Government representative upon completion of the production lot assembly. Loaded rocket catapult samples must not be preselected during production. Rocket catapults selected for samples shall not be included as a part of the quantity specified for delivery in the contract or order. Only primary components from a single primary component lot shall be used in a production lot of loaded rocket catapults. One primary component lot may be used in more than one loaded rocket catapult production lot. The ballistic samples must be selected as equally as possible among all rocket motor batches. From each rocket motor batch not represented in the sample rocket catapult assemblies, one sample rocket motor shall be selected for static firing by the contractor (see 4.5.5.1), so that at least one rocket motor from each rocket motor batch has been static fired.

Table IV. Sample size for firing test.

Lot size	Total firing sample	Sample per temperature extreme
Up to 110	10	5
111-180	14	7
181-300	20	10
301-340	22	11
341-380	24	12
381-420	26	13

4.5.2.4 Sampling for nondestructive test. Each rocket catapult asembly shall be subjected to the tests specified in 4.5.6.

4.5.3 Resubmission of rejected lot. A sample of rocket catapult assemblies shall be selected in accordance with 4.5.2.3 and submitted when, with prior approval of the procuring activity, the contractor requests that the rejected lot be test fired under different conditions or the contractor believes a rejected lot may be reworked to meet the requirements of this specification.

4.5.4 Examination.

4.5.4.1 Critical and major defects. Each rocket catapult assembly shall be visually and radiographically examined for critical and major defects specified in 4.5.2.2 and applicable drawings. Any rocket catapult assembly found to contain a critical or major defect shall be rejected.

4.5.4.2 Minor defects. Sample rocket catapult assemblies selected in accordance with 4.5.2.2.b shall be examined for minor defects in accordance with the applicable drawings, 3.8, 3.11, and 3.12. The acceptable quality level (AQL) shall be 2.5 defects per hundred units.

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4.5.5 Production lot acceptance firing. Rocket catapults selected in accordance with 4.5.2.3 shall be static fired at the Naval Ordnance Station, Indian Head, Maryland 20640-5000 within 30 days after submission for acceptance of the rocket catapult lot. The sample rocket catapults shall be divided into two equal groups. One group shall be conditioned at −65 °F and one at +165 °F for a minimum of 8 hours. The rocket catapults shall be fired within 10 minutes after removal from the conditioning chamber. Minimum equipment accuracies shall be as shown in table V. Data specified in table I shall be analyzed in accordance with MIL-STD-414, Section B, Part I, using the procedures of B-5 through B-7. Nonconformance with table I or any requirements of 3.4 shall be cause for rejection of the lot.

Table V. Test equipment accuracies.

Requirement	Equipment accuracy
Maximum thrust	±2.0 percent
Time	±0.001 second
Maximum acceleration	±5.0 percent
Pressure	±2.0 percent
Temperature	±2.0 °F

4.5.5.1 Contractor batch acceptance test firing. Rocket motors selected by the contractor to meet the requirements of 4.5.2.3 shall be static fired by the contractor as rocket motor batch acceptance tests. The sample rocket motors shall be divided into two groups as equal as possible. One group shall be conditioned at −65 °F and the other at +165 °F for a minimum of 8 hours. The rocket motors shall be fired within 10 minutes after removal from conditioning chamber. Minimum equipment accuracies shall be as shown in table V. Failure of any rocket motor to meet the requirements of 3.4 shall be cause for rejection of the specific batch of rocket motors.

4.5.5.2 Postfiring disassembly. A postfiring disassembly and inspection shall be conducted on all first article, lot acceptance, and batch acceptance test units. Detailed observations on conditions, including marginal conditions, of individual components shall be noted. All findings shall be evaluated against the requirements of 3.4.1.1. When specified in the contract or order (see 6.2.1), a disassembly findings report (see 6.2.2) shall be submitted to the contracting activity.

4.5.6 Nondestructive tests. Each rocket catapult assembly shall be subjected to the following tests. When specified in the contract or order (see 6.2.1), the contractor shall furnish test reports (see 6.2.2) detailing the results of all non-destructive tests for each lot of rocket catapults.

4.5.6.1 Radiographic inspection. Rocket catapults shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket catapult components 0.03 inch in length or width, whichever is smaller. When specified in the contract (see 6.2.1), all radiographs taken of the rocket catapult assembly or components thereof shall be forwarded to the activity designated by the contracting activity. Otherwise, all radiographs shall be inspected on-site by a designated Government representative. Acceptance of the loaded rocket catapult lot shall be subject to approval of the radiographs by the designated Government activity.

4.5.6.2 Hydrostatic pressure. Each component listed shall be subjected to the respective internal hydrostatic pressure specified in table II for a minimum of 15 seconds without application of an external load. Any leakage, permanent deformation, or mechanical failure shall result in rejection of that component. Each hydrostatically tested component shall then be subjected to magnetic particle inspection per 4.5.6.3.

4.5.6.3 Magnetic particle inspection. Each motor tube, booster tube, and launcher tube shall be subjected to magnetic particle inspected in accordance with MIL-I-6868. Any defect shall result in rejection of the defective part.

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4.5.6.4 Radiographic inspection. Each breech assembly shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. Any deficiency shall be cause for rejection of the assembly.

4.5.6.5 Penetrant inspection. Each breech, prior to anodizing, shall be subjected to penetrant inspection in accordance with ASTM E-165-80. Linear indications are unacceptable. Rounded penetrant indications, limited to 1/64-inch diameter, are permissible, with a maximum of 15 such indications per square inch allowed.

4.5.6.6 Torque. After ascertaining that each booster tube and breech assembly has been seated with 400 +50, −0 inch-pounds of torque, a break-away torque of 200 inch-pounds shall be applied. Any evidence of loosening or deformation of the booster tube and breech assembly shall result in rejection of the part.

4.5.6.7 Residual magnetism. Each rocket catapult assembly shall be passed 6 inches in front of an MS 17983 compass along the entire length of the catapult in accordance with figure 1. If the compass indicator deflects more than 5 degrees in either direction, the rocket catapult assembly shall be rejected.

4.6 Packaging inspection. The packaging, packing, and marking shall be inspected to verify conformance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or order (see 6.2.1), preservation, packaging, and packing shall be level A.

5.1.1 Level A. Preservation, packaging, and packing shall be in accordance with drawings listed on DL 5184322 and MIL-P-116. Containers shall conform to Uniform Freight Classification, National Motor Freight Classification, or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special markings required by the contract or order (see 6.2.1), each container shall be marked in accordance with MIL-STD-129, drawings listed on DL 5184322, and Code of Federal Regulations 49 CFR 171-178 (see 6.7.1).

6. NOTES

6.1 Intended use. The CKU-5/A/A Aircraft Ejection Seat Catapult Assembly is intended to be used to propel an aircraft ejection seat and its occupant safely out and away from a jet aircraft.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Quantity required (exclusive of first article and static firing test samples) (see 4.4 and 4.5.2.3).
c. When a first article sample is not required (see 3.1, 4.4, and 6.4).
d. Instructions for adaption of first article samples for pressure transducers (see 3.1.1 and 6.5).
e. Assigned activity for first article inspection and static firing tests (see 4.4, 4.4.2, 4.5.2.3, 4.5.5,and 6.6).
f. Whether the gun fire vibration test is required (see 4.4.1.2).
g. Whether test reports are required (see 4.5.1, 4.5.5.2, and 4.5.6).
h. When radiographs are to be furnished to the Government (see 4.5.6.1).
i. Packaging requirements if other than level A (see 5.1).
j. Any special markings required (see 5.2).
k. Safety precautions (see 6.7).
l. Disposition of hardware, scrap propellant and metal parts (see 6.8).
m. Inspection conditions if other than as specified (see 4.3).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.
4.5.1, 4.5.5.2, 4.5.6	Reports, Test	DI-T-2072
4.5.1	Ammunition Data Cards	DI-MISC-80043
4.5.1	Certification Data/Report	UDI-A-23264B

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Rocket motor resultant impulse. The rocket motor resultant impulse is obtained by vectorially adding the longitudinal and normal impulse of the rocket motor taken over the action time.

6.3.2 Rocket motor action time. Action time is determined from the normal thrust time curve, and shall be defined as the time interval from the 10 percent of maximum thrust on the initial rise of curve to the corresponding 10 percent of thrust on the declining portion of the curve.

6.3.3 Rocket motor resultant thrust. The rocket motor resultant thrust is obtained by vectorially adding the two maximum components of rocket thrust that occur at one time.

6.3.4 Rocket motor ignition delay time. Rocket motor ignition delay time is the elapsed time from separation (the point of the catapult thrust trace where the trace begins to drop sharply) to 10 percent of maximum thrust, as measured on the rising portion of the normal component rocket thrust trace.

6.3.5 Catapult ignition delay time. Catapult ignition delay time is the elasped time from the first indication of shearing of the firing pin shear pins to first indication of catapult thrust.

6.3.6 Catapult separation velocity. Catapult separation velocity is defined as the average velocity of the simulated ejected weight propelled horizontally at catapult separation measured over the interval from separation to 12 inches after separation.

6.3.7 Catapult dynamic response index - (DRI). DRI is representative of the maximum dynamic compression of the vertebral column of the human body, as defined in MIL-S-9479.

6.3.8 Propellant batch. A propellant batch is a mix of ingredients, each ingredient composed of one homogenous lot, that is mixed at one time in one vessel, resulting in a homogenous mixture.

6.3.9 Rocket motor batch. A rocket motor batch is a group of rocket motors cast from one propellant batch.

6.3.10 Rocket motor lot. A rocket motor lot is a group of rocket motors composed of one or more rocket motor batches all having the same ingredients lots.

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6.3.11 Rocket catapult lot. A rocket catapult lot is a group of units composed of one lot of rocket motors and one lot of catapult cartridges.

6.3.12 Primary components. Primary components are all components containing explosive ingredients. This includes the cartridge percussion primers, the cartridge, the rocket motor, and the rocket motor igniter(s).

6.4 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of 18 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Pressure transducers. Instructions for adaption of first article catapult assemblies for pressure transducers should be compatible with instrumentation that has a response of not less than 600 Hz and that is capable of detecting ignition peak pressures and unstable burning rates.

6.6 Test facility. The Naval Ordnance Station, Indian Head, Maryland is the Design Agent for the product covered by this specification. As such they are equipped to perform the first article inspection and static firing tests specified herein. Except when special conditions warrant otherwise, the Naval Ordnance Station, Indian Head should be designated as the test facility for the first article inspection and the production lot acceptance test firings (see 6.2.1(e)).

6.7 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M, are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE

When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, should be made applicable.

6.7.1 Explosive hazard classification. The explosive hazard classification for this rocket catapult has been designated as:

Explosive Hazard Classification - Class 1 Division 3
Storage Compatibility Group - Group C
Department of Transportation (DOT) Hazard Class - Cl B
DOT Marking - Propellant Explosive

6.7.2 Hazard notice. The rocket catapult described herein is flammable and/or explosive and consequently presents a hazard in manufacture, handling, storage, and shipment. The contractor should recognize this hazard and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidence.

6.8 Disposition of hardware, scrap propellant, and metal parts. The contract should specify that all items purchased become the property of the U.S. Government; no hardware, propellant, or metal parts acceptable, scrapped, or tested be disposed of without authorization of the contracting activity; all items should be made available to the contracting activity upon request; and that the disposal of any tested hardware, propellant, or metal parts without authorization by the contracting activity will result in the test being declared invalid, and a retest will be required.

6.9 Retention of radiographs. All radiographs of accepted lots of rocket catapults are to be retained on file by the contractor for 9 years or 1 year past the service life of the unit, whichever is longer.

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6.10 Life. The rocket catapult has a design life of 8 years with an installed service life of 8 years from the date of propellant manufacture.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.12 Subject terms (keyword) listing.

Ejection equipment, aircraft
Propellant
Propellant actuated device
Safety equipment, aircrew

Preparing Activity
Navy - OS
(Project 1377-N931)

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